
Fused-Regime Student- t Dynamic Bayesian Networks for Cross-Asset Dependency Structure

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Abstract

We extend DYNOTEARS, a continuous-optimization learner of dynamic Bayesian networks (DBNs), along two axes motivated by financial time series: a fused multi-regime penalty, whose scientific object is the sparse *change graph* $\Delta_W = W^{\text{event}} - W^{\text{ordinary}}$, and a Student- t likelihood for heavy-tailed residuals. The composite objective combines a smooth heavy-tailed loss, a nonsmooth sparsity-plus-fusion penalty, and the nonconvex NOTEARS acyclicity constraint. We solve it with consensus ADMM, applying a closed-form $K=2$ fused-lasso proximal step inside an augmented-Lagrangian outer loop. This is the directed, regime-conditioned, heavy-tailed analogue of the fused graphical lasso. On a var-sortability-controlled synthetic benchmark (designs standardized so var-sortability = 0.50), the full estimator recovers the change graph +0.19 to +0.31 change- W AUROC better than the DYNOTEARS-equivalent across change-edge count, sample size, and tail heaviness, with the gap widest under heavy tails. The ADMM/exact-prox implementation is the largest observed lever, Student- t and fusion help secondarily, and held-out-likelihood selection chooses fusion ($\gamma > 0$) in every seed. We then apply the method to a $d = 23$ hourly cross-asset panel (2024–2026) around U.S. macro releases (CPI/NFP/FOMC). The contemporaneous DAG is strongly justified (out-of-sample $W \equiv 0$ gate: DAG $\gg$ SVAR; Student- t beats Gaussian density), but the empirical hypothesis is *not* supported for the contemporaneous change graph: global and edge-wise tests are null. Lagged edge-wise tests also have no BH-FDR discoveries; the lone lagged global rejection reflects event-regime shrinkage rather than interpretable edge-level reorganization. Held-out selection independently drives the fusion penalty $\gamma \rightarrow 0$. We report this as a carefully-controlled negative result, separate from the validated methodological contribution.

1 Introduction

DYNOTEARS [Pamfil et al., 2020] learns dynamic Bayesian networks (DBNs) from time series. It assumes a single, stationary series. Its key idea is to recast the combinatorial search over directed acyclic graphs as continuous optimization, using the smooth NOTEARS acyclicity surrogate [Zheng et al., 2018]. The estimator is elegant and widely used. Two settings it handles poorly, however, are exactly the ones a financial application forces on us. First, heterogeneous regimes: when calendar information partitions the data into states (e.g. ordinary trading versus a scheduled macroeconomic event) that share most of their dependency structure, fitting one graph blurs the regimes, while fitting independent graphs wastes the shared signal, especially when one regime is small. Second, heavy tails: asset returns have fat tails, so the Gaussian-equivalent squared loss implicit in DYNOTEARS is a mismatched likelihood that over-weights large innovations.

*Code and data: <https://github.com/0xknxwledge/TTIC31180FinalProject>

We propose FR-tDBN (Fused-Regime Student- t DBN), which augments DYNOTEARS with two ingredients. First, a fused penalty $\gamma \|\mathbf{W}^1 - \mathbf{W}^0\|_1$ across $K=2$ regimes, so edges that do not change pay no cost and the small event regime borrows strength from the large ordinary regime; the estimand is the sparse change graph $\Delta_W = \mathbf{W}^1 - \mathbf{W}^0$. Second, a fixed- ν Student- t data term in place of squared loss. The fused penalty does not produce a sparse Δ_W under the smoothed- ℓ_1 surrogate that NOTEARS-style solvers use, because $\sqrt{x^2 + \epsilon}$ never reaches exactly zero. We therefore solve the true composite objective with consensus ADMM, applying a closed-form fused-lasso proximal operator (the $K=2$ case of Friedman et al., 2007, Danaher et al., 2014) to the nonsmooth penalties and L-BFGS to the smooth data-plus-acyclicity part, inside an augmented-Lagrangian loop for the nonconvex constraint. This is the directed, regime-conditioned, heavy-tailed analogue of the fused/joint graphical lasso [Danaher et al., 2014].

Because real financial graphs have no ground-truth DAG, the methodological claim is earned on simulation. Our synthetic benchmark explicitly controls for *var-sortability* [Reisach et al., 2021]: the generator is var-sortable by construction, so we standardize each regime before fitting, pinning measured var-sortability at 0.50 and removing the variance gradient that can make NOTEARS-style methods look good for the wrong reason. On this benchmark FR-tDBN clearly out-recovers the DYNOTEARS-equivalent, and the win survives a distribution-free rank transform.

We then apply the method to a small but complete $d = 23$ crypto-macro panel sampled hourly over two years, asking a concrete empirical question: *does the conditional dependency structure linking crypto, equities, rates, FX, and volatility reorganize around scheduled macro releases (CPI/NFP/FOMC), and which edges flip?* The contemporaneous DAG is strongly justified. But tested against a volatility-matched null, the contemporaneous change graph is indistinguishable from chance at both global and edge-wise resolution. Lagged edge-wise tests likewise find no BH-FDR discoveries; the small global lagged-norm rejection is a diagnostic of event-regime shrinkage, not a stable set of changing lead-lag edges. Principled model selection corroborates the contemporaneous null independently by driving the fusion penalty to zero. We report the null as the defensible empirical statement.

Contributions.

1. **Method.** A fused multi-regime, Student- t DBN whose estimand is a sparse change graph, with a consensus-ADMM solver that applies an exact closed-form fused-lasso prox inside the NOTEARS augmented-Lagrangian loop (§2).
2. **Var-sortability-controlled evidence.** On a standardized benchmark, FR-tDBN recovers the change graph +0.19 to +0.31 AUROC better than the DYNOTEARS-equivalent across the design space; a one-codebase ablation associates the gain mainly with the ADMM/exact-prox implementation, then the Student- t loss and fusion, and confirms robustness to a rank transform and to principled (λ, γ) selection (§3).
3. **A controlled cross-asset null.** On real data we justify the contemporaneous DAG out-of-sample, then show with volatility-matched global and edge-wise permutation tests (and corroborating held-out selection) that scheduled macro events do *not* measurably reorganize the contemporaneous graph at hourly resolution, while lagged edge-wise tests find no stable event-specific edges (§4).

2 Method: Fused-Regime Student- t DBN

2.1 Model and notation

Regimes $k \in \{0, 1\}$ (0 = ordinary, 1 = event); d variables; lag order p ; n_k observations in regime k ; $N = n_0 + n_1$. Per regime the lagged design comprises contemporaneous targets $\mathbf{X}^k \in \mathbb{R}^{n_k \times d}$ and lagged copies $\mathbf{Y}_\ell^k \in \mathbb{R}^{n_k \times d}$ for $\ell = 1, \dots, p$. Parameters $\Theta = (\mathbf{W}^0, \mathbf{W}^1, \mathbf{A}_{1:p}^0, \mathbf{A}_{1:p}^1)$ consist of per-regime intra-slice DAGs $\mathbf{W}^k \in \mathbb{R}^{d \times d}$ ($\text{diag}(\mathbf{W}^k) = 0$; entry $\mathbf{W}^k[i, j]$ is the directed edge $i \rightarrow j$, so column j is the structural equation for variable j) and inter-slice matrices $\mathbf{A}_\ell^k$. The residuals are

$$\mathbf{E}^k = \mathbf{X}^k - \mathbf{X}^k \mathbf{W}^k - \sum_{\ell=1}^p \mathbf{Y}_\ell^k \mathbf{A}_\ell^k \quad (n_k \times d), \quad (1)$$

modeled i.i.d. Student- t with shared, fixed degrees of freedom ν and fixed per-variable scales σ_j (pooled robust MAD).

2.2 Objective

We minimize $F(\Theta) = f(\Theta) + g(\Theta)$ subject to per-regime acyclicity. The smooth data term is normalized per observation so penalties transfer across n (the larger ordinary regime carries proportionally more weight, which is the borrow-strength mechanism):

$$f(\Theta) = \frac{1}{N} \sum_{k \in \{0,1\}} \sum_{t=1}^{n_k} \sum_{j=1}^d \ell(\mathbf{E}^k[t, j]; \nu, \sigma_j), \quad (2)$$

with, at fixed ν, σ (additive ν -constants dropped for training; restored for held-out density, §4.2),

$$\ell_t(e; \nu, \sigma) = \log \sigma + \frac{\nu+1}{2} \log \left(1 + \frac{e^2}{\nu \sigma^2} \right), \quad \ell_N(e; \sigma) = \log \sigma + \frac{e^2}{2\sigma^2}. \quad (3)$$

The nonsmooth penalty combines within-regime sparsity and cross-regime fusion,

$$\begin{aligned} g(\Theta) = & \lambda_W (\|\mathbf{W}^0\|_1 + \|\mathbf{W}^1\|_1) + \lambda_A \sum_{\ell} (\|\mathbf{A}_{\ell}^0\|_1 + \|\mathbf{A}_{\ell}^1\|_1) \\ & + \underbrace{\gamma_W \|\mathbf{W}^1 - \mathbf{W}^0\|_1}_{\text{fusion}} + \gamma_A \sum_{\ell} \|\mathbf{A}_{\ell}^1 - \mathbf{A}_{\ell}^0\|_1, \end{aligned} \quad (4)$$

subject to $h(\mathbf{W}^k) = 0$ and $\text{diag}(\mathbf{W}^k) = 0$, where $h(\mathbf{W}) = \text{tr}(e^{\mathbf{W} \circ \mathbf{W}}) - d$ is the NOTEARS surrogate ($\circ$ = Hadamard product). Here f is smooth, g is convex but nonsmooth, and h is smooth but nonconvex; the fusion term is the heart of the model, since unchanged edges pay no cost and the reported object is the sparse $\Delta_W = \mathbf{W}^1 - \mathbf{W}^0$.

2.3 Optimization

We separate the three term types: L-BFGS for the smooth part, an augmented-Lagrangian outer loop for the nonconvex acyclicity, and a closed-form proximal operator for the nonsmooth penalties via consensus ADMM [Boyd et al., 2011]. The idea of ADMM is to give the smooth and the nonsmooth terms their own copy of the parameters, let each subproblem solve the piece it is suited to (gradients for the smooth loss, thresholding for the penalties), then reconcile the copies. Concretely, introduce penalty copies $\mathbf{Z} = (\mathbf{V}^0, \mathbf{V}^1, \mathbf{B}_{1:p}^0, \mathbf{B}_{1:p}^1)$ of Θ under the consensus constraint $\Theta = \mathbf{Z}$, together with a scaled dual variable $\mathbf{U}$ that accumulates the running disagreement $\Theta - \mathbf{Z}$ and is what forces the two copies to exact equality at a finite penalty ρ . The smooth subproblem the L-BFGS step sees is

$$S(\Theta) = f(\Theta) + \sum_k \left[\frac{\rho_k}{2} h(\mathbf{W}^k)^2 + \alpha_k h(\mathbf{W}^k) \right]. \quad (5)$$

The Θ -update is smooth (the only nonconvexity is h^2), so L-BFGS is well-behaved; all sparsity and fusion live in the $\mathbf{Z}$ -update. A proximal operator $\text{prox}_h(a) = \arg \min_v h(v) + \frac{1}{2} \|v - a\|^2$ returns the point that lowers a penalty h while staying near its input a ; for an ℓ_1 penalty it is soft-thresholding, which shrinks each coordinate toward zero and snaps small ones to *exactly* zero (the source of sparsity). Our $K=2$ fused-lasso prox [Tibshirani et al., 2005, Friedman et al., 2007] applies two such pulls at once: each weight is pulled small (ℓ_1) and the two regimes are pulled equal (fusion). For each coordinate, FUSEDPROX2($a^0, a^1; \lambda', \gamma'$) solves $\min_v \frac{1}{2}(v^0 - a^0)^2 + \frac{1}{2}(v^1 - a^1)^2 + \lambda'(|v^0| + |v^1|) + \gamma'|v^1 - v^0|$ by *fuse, then soft-threshold*. The fuse step (preserving the mean $(a^0 + a^1)/2$) is

$$(t^0, t^1) = \begin{cases} (a^0 + \gamma', a^1 - \gamma') & \text{if } a^1 - a^0 > 2\gamma' \\ (a^0 - \gamma', a^1 + \gamma') & \text{if } a^1 - a^0 < -2\gamma' \\ ((a^0 + a^1)/2, (a^0 + a^1)/2) & \text{otherwise,} \end{cases} \quad (6)$$

followed by the soft-threshold $v^k = \text{sign}(t^k) \max(|t^k| - \lambda', 0)$. The difference is soft-thresholded by $2\gamma'$, forcing $\mathbf{W}^0 = \mathbf{W}^1$ (hence exact zeros in Δ_W) on unchanged edges, which is the selection behavior smoothed- ℓ_1 lacks. Empirically $\max_k h(\mathbf{W}^k) \approx 4 \times 10^{-3}$ at convergence, and ADMM $\times$ nonconvex acyclicity did not require the warm-start fallback we had budgeted for.

Algorithm 1 FR-tDBN: consensus ADMM inside the acyclicity augmented Lagrangian

```
1: init  $\Theta \sim$  small random ( $\text{diag } \mathbf{W}=0$ );  $\mathbf{Z} \leftarrow \Theta$ ;  $\mathbf{U} \leftarrow 0$ ;  $\alpha \leftarrow 0$ ;  $\rho_h \leftarrow 1$ ;  $h_{\text{prev}} \leftarrow \infty$ 
2: for outer  $m = 1, \dots, M$  do ▷ acyclicity augmented Lagrangian
3:   for  $t = 1, \dots, T_{\text{admm}}$  do ▷ consensus ADMM
4:      $\Theta \leftarrow \arg \min_{\Theta} S(\Theta) + \frac{\rho}{2} \|\Theta - \mathbf{Z} + \mathbf{U}\|_F^2$  ▷ smooth: L-BFGS, strong-Wolfe
5:      $\mathbf{Z} \leftarrow \text{FUSEDPROX2}(\Theta + \mathbf{U}; \lambda/\rho, \gamma/\rho)$  ▷ closed-form, exact zeros;  $\text{diag } \mathbf{V}=0$ 
6:      $\mathbf{U} \leftarrow \mathbf{U} + \Theta - \mathbf{Z}$  ▷ dual update
7:   end for
8:    $h_{\text{max}} \leftarrow \max_k |h(\mathbf{W}_{\Theta}^k)|$ 
9:   if  $h_{\text{max}} > 0.25 h_{\text{prev}}$  and  $\rho_h < \rho_h^{\text{max}}$  then
10:     $\rho_h \leftarrow 10 \rho_h$ 
11:   else
12:     $\alpha_k \leftarrow \alpha_k + \rho_h h(\mathbf{W}_{\Theta}^k)$ ;  $h_{\text{prev}} \leftarrow h_{\text{max}}$ 
13:   end if
14:   if  $h_{\text{max}} \leq h_{\text{tol}}$  break
15: end for
16: report  $\mathbf{W}^k \leftarrow \mathbf{V}^k$  (exactly sparse),  $\mathbf{A}_{\ell}^k \leftarrow \mathbf{B}_{\ell}^k$ ,  $\Delta_{\mathbf{W}} = \mathbf{V}^1 - \mathbf{V}^0$ ,  $\Delta_{\mathbf{A}} = \mathbf{B}^1 - \mathbf{B}^0$ 
```

2.4 Variants and ablations

A single codebase exposes four switches for comparative attribution. The loss $\in \{\text{Student-}t, \text{Gaussian}\}$ isolates heavy tails; the fusion strength toggles $\gamma=0$ (independent per-regime fits) against $\gamma>0$; the solver $\in \{\text{admm}$ (exact prox, headline), `lbfgs_smooth` (smoothed $\sqrt{x^2 + \epsilon}$, the DYNOTEARS-style baseline) $\}$ compares exact-prox ADMM to the smoothed- ℓ_1 implementation; and an adaptive-fusion option sets $\gamma_{ij} = \gamma_0 / (|\hat{\Delta}_{ij}| + \epsilon)$. Because the smooth baseline normalizes its smoothed penalties with the loss while ADMM applies the nonsmooth prox at the stated λ, γ scale, the solver arm is a practical baseline comparison rather than a perfectly scale-matched prox-only ablation. The Gaussian, independent, smoothed- ℓ_1 corner is our reimplement of DYNOTEARS, labeled the “DYNOTEARS-equivalent” throughout.

3 Synthetic benchmark

Generator and var-sortability control. `make_regime_pair` samples an Erdős–Rényi ordinary DAG, stabilizes the reduced-form VAR, then applies a single stable step to a sparse delta (additions, removals, reweights), so the change masks equal the realized $\mathbf{W}^1 - \mathbf{W}^0$ and $\mathbf{A}^1 - \mathbf{A}^0$ exactly (no rescale contamination), $\Delta_{\mathbf{W}}$ stays sparse, and regime sizes can be imbalanced ($n_{\text{event}} \ll n_{\text{ordinary}}$). The generator is var-sortable by construction (variance grows down the topological order; raw score ≈ 0.83 at $\nu=5$), so we *standardize each regime* before fitting. This pins measured var-sortability at exactly 0.50 [Reisach et al., 2021], so recovery cannot ride the variance gradient and the result transfers to the standardized real pipeline. We score change-edge AUROC, the ranking of $|\widehat{\Delta_{W_{ij}}}|$ against the true change support.

Headline grid. Table 1 reports change- $\mathbf{W}$ AUROC over a solver $\times$ loss $\times$ fusion grid, 24 fits per cell. Three findings are robust at $\gtrsim 2$ s.e. in both the balanced/moderate-tail regime ($\nu=5$, $n \in \{50, 100\}$) and the small-sample/heavy-tail regime ($\nu=3$, $n \in \{20, 35\}$): (i) the ADMM/exact-prox implementation outperforms smoothed- ℓ_1 everywhere (+0.13 or more), and even ADMM-independent beats smooth-fused, so this is the largest observed lever, subject to the penalty-scaling caveat above; (ii) Student- t helps under ADMM (+0.10 to 0.12 over Gaussian); and (iii) fusion helps the weaker (Gaussian) estimator (+0.06 to 0.08) and the balanced regime, while its marginal gain on top of Student- t is within noise. The full FR-tDBN (ADMM, Student- t , fused) is the best or statistically-tied-best cell in both regimes.

Sweeps across the design space. Figure 1 sweeps one axis at a time (12 seeds), comparing the full FR-tDBN against the DYNOTEARS-equivalent. The gap is large and stable: change-edge count $1 \rightarrow 5$, +0.25 to +0.30; sample size $50 \rightarrow 200$, +0.27 to +0.30; and tail heaviness $\nu 3 \rightarrow 30$, +0.31 (heavy) down to +0.19 ($\approx$ Gaussian). The gap is *largest under heavy tails*, exactly as the Student- t likelihood

solver	loss	$\nu = 5, n \in \{50, 100\}$		$\nu = 3, n \in \{20, 35\}$	
		independent	fused	independent	fused
admm	Gaussian	0.714 (.030)	0.794 (.032)	0.565 (.033)	0.626 (.027)
admm	Student- t	0.838 (.027)	0.853 (.031)	0.666 (.033)	0.662 (.027)
lbfgs_smooth	Gaussian	0.584 (.033)	0.582 (.033)	0.538 (.031)	0.562 (.032)
lbfgs_smooth	Student- t	0.617 (.033)	0.626 (.031)	0.524 (.029)	0.537 (.032)

Table 1: Synthetic change- W AUROC, mean (s.e.) over 24 fits/cell, designs standardized so var-sortability = 0.50. Best per block in **bold**.

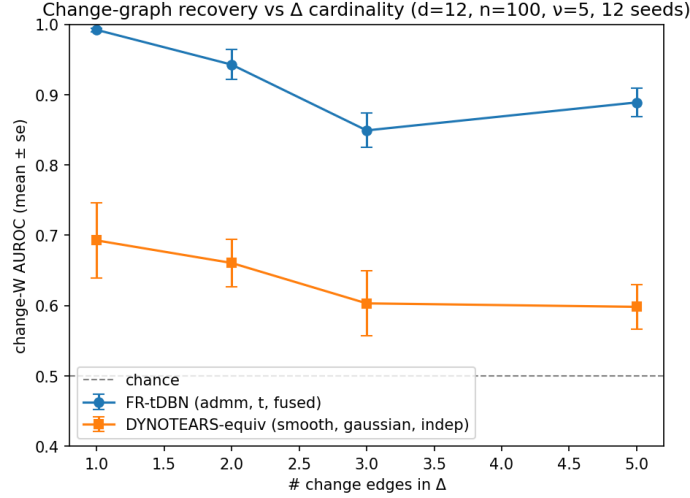


Figure 1: Change-graph recovery vs. Δ_W cardinality ($d=12, n=100, \nu=5, 12$ seeds). FR-tDBN (ADMM, Student- t , fused) recovers the change support far above the DYNOTEARS-equivalent (smoothed- ℓ_1 , Gaussian, independent) across the sweep; both stay above chance (0.5). The advantage is reproduced over sample-size and tail-heaviness sweeps (text).

predicts. A change-edges = 0 control gives a clean Δ_W -floor ($\|\widehat{\Delta}_W\|_1 \approx 0.27$), confirming the estimator does not hallucinate change where there is none.

Robustness of the methodological claim. Two checks guard against artifacts. (a) Rank transform. Replacing standardization with a distribution-free rank (Gaussian-copula) transform shrinks the FR-tDBN advantage from +0.24 to +0.14 change- W AUROC but leaves it clearly positive. The shrinkage is expected, since ranking compresses the heavy tails the Student- t exploits, so the win is not a marginal-distribution artifact. (b) Principled (λ, γ) selection. Selecting penalties by held-out NLL on ground-truth data chooses fusion ($\gamma=0.02 > 0$) in 5/5 seeds (modal $\lambda=0.03$); BIC is more conservative ($\lambda=0.1$, small γ). Fusion is thus chosen by a predictive criterion, not hand-tuned. We note that the held-out λ sits at the grid’s low edge while BIC’s sits high, so the scientific parameter γ is the stable, interpretable one. (c) Loss form. Swapping the Student- t data term for a pseudo-Huber or Normal-Inverse-Gaussian loss leaves the recovery advantage essentially unchanged (App. A), so the gain reflects robustness to heavy tails rather than the specific Student- t form.

A documented negative ablation. Adaptive fusion ($\gamma_{ij} = \gamma_0 / (|\hat{\Delta}_{ij}| + \epsilon)$, in the spirit of Zou, 2006, Candès et al., 2008) *fails* here: at small event- n every available pilot already shrinks the true Δ , so the reweighting over-penalizes the genuine change edges and collapses Δ_W to near-empty. The unbiased-pilot assumption is violated; we report the failure rather than omit it. A related limitation is that exact support recovery (F1) at small event- n is hard even for uniform fusion (mean F1 0.15 across $n \in \{20, 35\}$, 0.20 at $n = 35$), while adaptive fusion falls to F1 0.06–0.09 with near-empty supports. This is why we report AUROC and adopt stability selection on real data rather than trusting a single fit’s support.

block	members
equity / crypto-equity	SPY, QQQ, IWM, NVDA, COIN, MSTR
rates / credit	TLT, IEF, SHY, HYG
commodities	GLD, SLV, USO
FX	DXY, EURUSD, GBPUSD, JPY
volatility	VIX
crypto (spot)	BTC, ETH, SOL, XRP, LINK

Table 2: The $d = 23$ cross-asset panel; crypto is one of the six blocks.

4 Cross-Asset Structure Around U.S. Macroeconomic Releases

We now ask whether the conditional dependency structure linking equities, rates, FX, commodities, volatility, and crypto reorganizes around scheduled U.S. macro releases (CPI/NFP/FOMC). The panel is deliberately cross-asset: crypto is one block among six. We give the crypto names modest extra attention because their returns are the most heavily tailed in the panel, which is exactly where the Student- t likelihood earns its keep, but the question and the panel are macro-wide rather than crypto-specific.

4.1 Data and event regime

We use hourly close prices from Yahoo Finance over the common window 2024-05-30 to 2026-05-28 (UTC). The aligned return panel contains 2,962 hourly return rows across 498 trading dates before preprocessing. The $d = 23$ panel spans six blocks (Table 2). Construction floors each timestamp to the hour, aligns on the union grid, takes log returns, drops any-NaN rows, then applies a rolling past-only z -score (window 250, min-periods 60; no look-ahead). After the rolling warmup and $p = 1$ lagging, the fitted design contains 2,901 target rows across 488 dates. We use $p = 1$, justified empirically below.

The event regime comes from `data/events.csv`, which holds CPI/NFP/FOMC release times (08:30 ET for CPI/NFP, 14:00 ET for FOMC), verified against official BLS/Fed schedules including the 2025 lapse-driven exceptions, converted ET→UTC with DST awareness. Bars within ± 2 h of a release are tagged event, giving 103 event bars (3.6%; 22 CPI, 21 NFP, 60 FOMC): the small- n_{event} borrow-strength regime the fusion targets. Because 08:30-ET CPI/NFP precede the cash-equity open, their windows on this RTH panel capture the post-open reaction; FOMC at 14:00 ET is mid-session and captured directly.

4.2 Evaluation protocol

With no ground-truth DAG on real data, we rely on out-of-sample density and controlled nulls. We use a time-ordered train/test split (last 20% held out) with train-derived scales for every held-out number (no leakage), and a full-density NLL (restoring the Γ and $\frac{1}{2} \log(\nu\pi)$ constants) so that Student- t -vs-Gaussian and DAG-vs- $(W \equiv 0)$ comparisons are valid. The $W \equiv 0$ SVAR ablation is the acyclicity/frequency sanity check: only an out-of-sample win counts, since the DAG has strictly more parameters. We assess change edges by block-bootstrap stability selection [Meinshausen and Bühlmann, 2010] and test significance with global and edge-wise permutation nulls on both Δ_W and Δ_A that are block- and volatility-matched (event windows are mechanically high-volatility), with Benjamini–Hochberg FDR control [Benjamini and Hochberg, 1995] on the edge-wise tests.

4.3 Results

The contemporaneous DAG is real (OOS $W \equiv 0$ gate). Table 3 shows every DAG model beats the $W \equiv 0$ SVAR out-of-sample by a wide margin (roughly 4.4k nats for Student- t and 7k nats for Gaussian), so the hourly contemporaneous structure genuinely carries the dependency information. Student- t density beats Gaussian for every model, confirming heavy tails on real data. This also reads as explanatory power of the joint structure, not forecasting.

Recovery wins, held-out density does not (vs. DYNOTEARS). The same table draws the key distinction. On ground-truth *recovery* FR-tDBN beats the DYNOTEARS-equivalent (§3), but on

model	Student- t NLL	Gaussian NLL
FR-tDBN (admm, Student- t , fused)	13,839	14,738
DYNOTEARS-per-regime (smooth, Gaussian, indep)	13,676	14,456
DYNOTEARS-pooled (smooth, Gaussian, single graph)	13,637	14,381
SVAR ($W \equiv 0$)	18,287	21,573

Table 3: Out-of-sample test NLL (nats; lower is better) on the held-out final 20%. DAG models crush the $W \equiv 0$ SVAR; Student- t beats Gaussian throughout; among DAGs, pooled DYNOTEARS edges FR-tDBN on held-out density (the recovery advantage is on synthetic ground truth, Table 1). Best in **bold**.

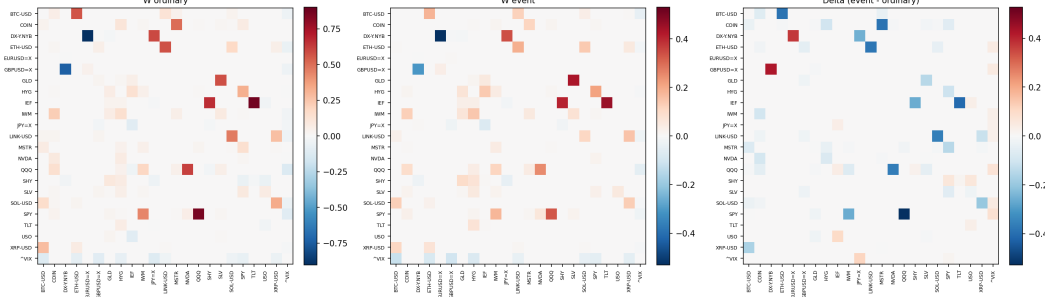


Figure 2: Estimated contemporaneous structure on the real panel: ordinary W^0 (left), event W^1 (middle), and the change graph $\Delta_W = W^1 - W^0$ (right). The two regimes share most structure; the fused estimator concentrates differences into a sparse Δ_W . The global magnitude of Δ_W , however, is not beyond a volatility-matched null (Table 4).

real-data held-out density it does not: pooled Gaussian DYNOTEARS is slightly better (roughly 160 to 360 nats), because the thin event regime gives fusion little to exploit out of sample. The defensible claim is better change-graph recovery on ground truth, not better held-out density on this panel.

Stable change edges are on-thesis but not event-specific. Across 25 block bootstraps the most stable Δ_W edges (frequency 1.0) are crypto-internal and cross-asset: ETH $\leftrightarrow$ SOL/LINK/XRP, BTC $\leftrightarrow$ XRP, BTC $\rightarrow$ VIX, NVDA $\rightarrow$ COIN, MSTR $\rightarrow$ SPY/QQQ, DXY $\rightarrow$ JPY (Fig. 2, right panel). The crypto block reliably reorganizes across bootstraps, but, as the nulls below show, this is sampling-stability, not event-specificity.

The contemporaneous change graph is null (global and edge-wise). This is the central empirical result. *Globally*, observed $\|\Delta_W\|_1 = 7.97$ against a volatility/block-matched regime-label null with mean 9.26 ($p \approx 0.95$): the magnitude of change is not beyond random vol-matched relabeling. *Edge-wise*, only 9 edges reach raw $p < 0.05$ versus ≈ 25 expected by chance, and 0 survive BH-FDR. The change is concentrated neither in magnitude nor in specific edges. The result is robust across event type and window (Table 4): every observed $\|\Delta_W\|_1$ is at or below its matched-null mean (all $p \geq 0.76$).

Structure is overwhelmingly contemporaneous, and lag order 1 is parsimonious. OOS NLL is flat across the checked lag orders $p \in \{1, 2, 3, 6\}$ (range 0.2%; $p = 6$ is lowest by only 0.18%) and lags ≥ 2 carry negligible weight ($\|A_\ell\|_1 \approx 0.01$ vs. $\|A_1\|_1 \approx 0.76$ vs. $\|W\|_1 \approx 14.8$). The hourly cross-asset dependency is dominantly contemporaneous; lead-lag is $\sim 20\times$ weaker and effectively zero beyond 1h, since genuine lead-lag lives at sub-hourly scales this two-year panel cannot resolve. The lag-1 graph still has a few sensible persistent edges, discussed with Fig. 5. Critically, the lagged structure has no edge-wise support around events: Δ_A gives 7 raw edges at $p < 0.05$ vs. ≈ 26 by chance and 0 BH survivors. The global $\|\Delta_A\|_1$ statistic is small but rejects (0.987 vs. null mean 0.942, $p \approx 0.040$ over 100 permutations); inspection shows this is a thin-event-regime artifact, since the 103-row event regime cannot estimate the already-small lead-lag graph, so $A^{\text{event}} \approx 0$ and $\Delta_A \approx -A^{\text{ordinary}}$. Thus W is null globally and edge-wise, and A has no edge-wise discoveries despite the global norm diagnostic.

setting	n_{event}	observed $\ \Delta_W\ _1$	null mean	p
type = FOMC	60	11.80	13.37	1.00
type = CPI	22	14.64	14.66	0.76
type = NFP	21	14.55	14.66	1.00
window $\pm 1\text{h}$	45	12.26	14.17	1.00
window $\pm 2\text{h}$	103	7.97	9.26	0.95
window $\pm 4\text{h}$	218	5.11	5.61	0.86

Table 4: Volatility-matched permutation null for $\|\Delta_W\|_1$ across event types and windows. Every observed norm is at or below its null mean; no setting rejects.

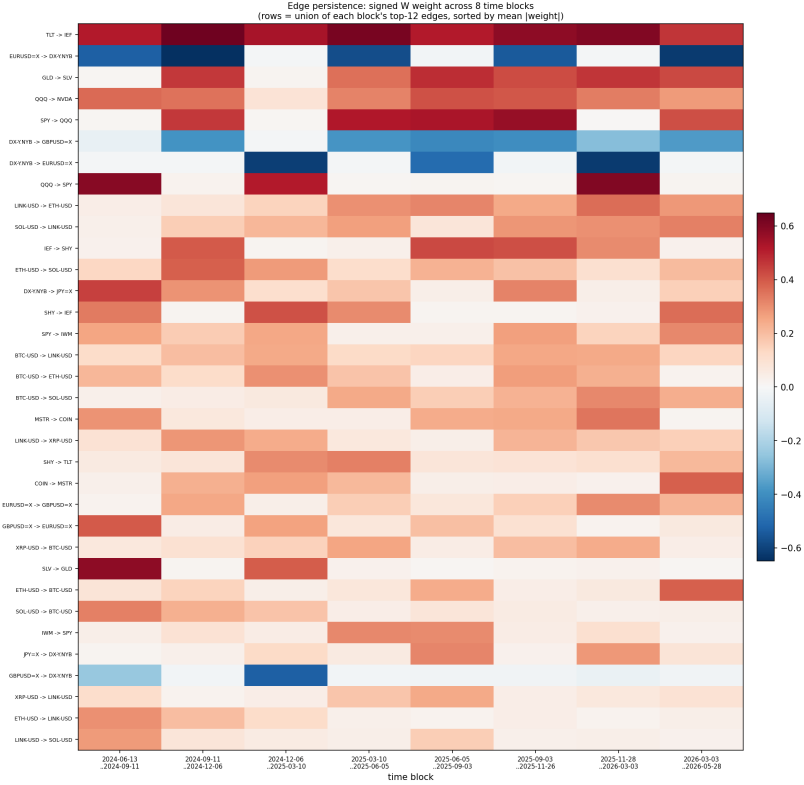


Figure 3: Signed contemporaneous edge weights across eight two-month blocks (rows = union of each block’s top-12 edges, sorted by mean $|weight|$; entry $W[i, j]$ is the $i \rightarrow j$ regression coefficient). The rates anchor $TLT \rightarrow IEF$, the dollar block ($EURUSD \rightarrow DXY$, $DXY \rightarrow GBPUSD$), metals ($GLD \rightarrow SLV$), and equity ($SPY \rightarrow QQQ$, $QQQ \rightarrow NVDA$) recur in nearly every block, with crypto links ($LINK \rightarrow ETH$, $SOL \rightarrow LINK$, $BTC \rightarrow ETH$) strengthening through 2025. The backbone’s stability argues against a sharp event-driven reorganization.

Persistent cross-asset structure (association, not cause). Fitting one graph per two-month block (Figs. 3–4) exposes a dependency backbone that recurs in essentially every block and splits cleanly by asset class: a Treasury-duration complex ($TLT/IEF/SHY$, with $TLT \rightarrow IEF$ the single most persistent edge), a broad-equity block ($SPY/QQQ/IWM/NVDA$), a metals pair (GLD/SLV), a dollar complex (DXY against $EUR/GBP/JPY$), and a tightly comoving crypto block ($BTC/ETH/SOL/LINK/XRP$) bridged to equities through $COIN$ and $MSTR$. We stress that these are *conditional associations*, not causal effects: standardizing away var-sortability removes the only cue that could orient contemporaneous edges, so we read each cluster as “these assets move together,” not “ X drives Y .” What is striking, and all we claim, is that the same structure persists across two years of very different market conditions.

Selection corroborates the null. Independently of the permutation tests, held-out-likelihood (λ, γ) selection on the real panel drives the fusion penalty to $\gamma \rightarrow 0$ (held-out NLL is minimized at $\gamma=0$;

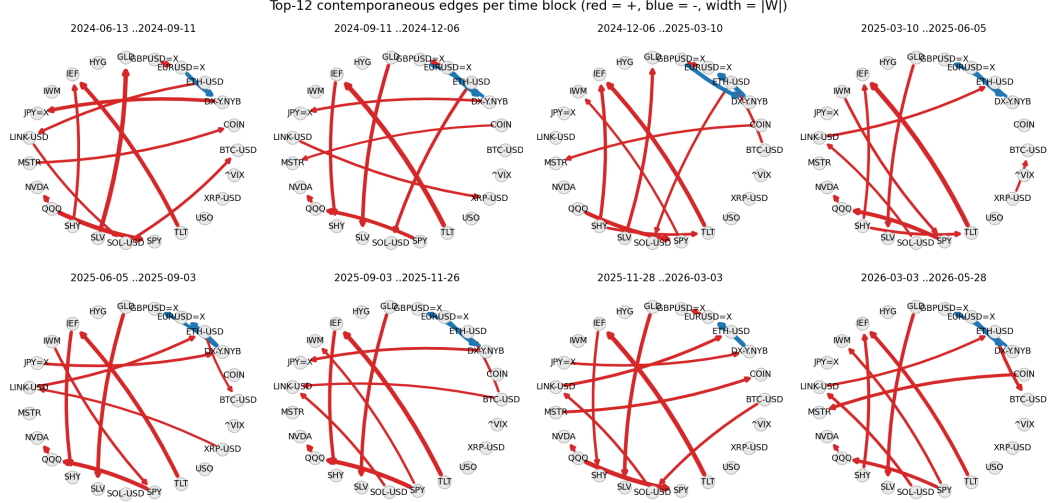


Figure 4: Top-12 contemporaneous edges per two-month block on a shared node layout and width scale (red = +, blue = -, width = $|W|$). The same blocks light up across time: the equity hub (SPY/QQQ/NVDA), the rates trio (TLT/IEF/SHY), the dollar/FX fan (DXY/EUR/GBP/JPY), metals (GLD/SLV), and the crypto cluster (BTC/ETH/SOL/LINK/XRP) with its COIN/MSTR equity bridges. Arrows are regression directions, not causal claims; the takeaway is the persistence of the cluster structure, not its orientation.

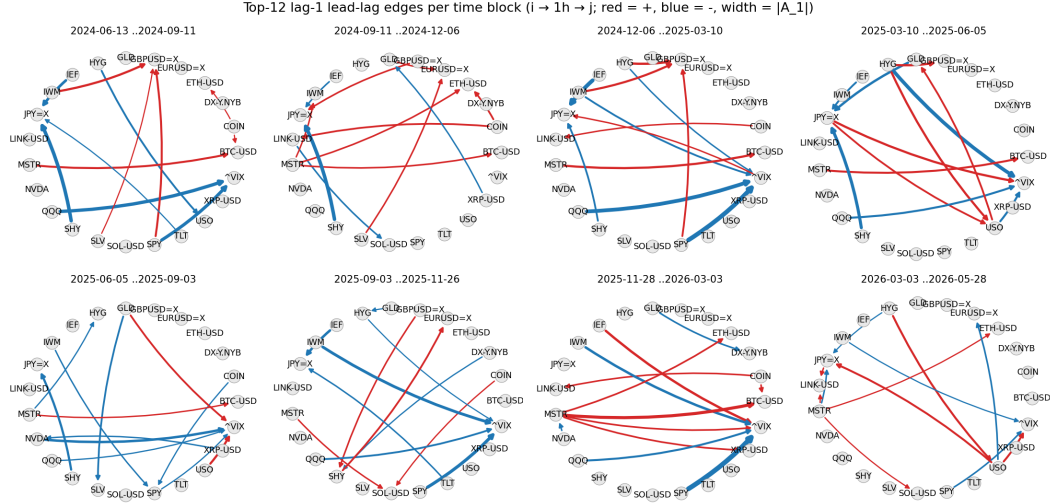


Figure 5: Top-12 lag-1 lead-lag edges per block (i at $t-1h \rightarrow j$ at t ; $= |A_1|$, $\sim 20\times$ weaker than W , note the smaller weights). A few edges recur: equities lead VIX down the next hour (SPY/QQQ/WTM $\rightarrow$ VIX, negative), front-end rates lead the yen (SHY/IEF $\rightarrow$ JPY), and the RTH-traded crypto proxies lead spot crypto (MSTR $\rightarrow$ BTC in six of eight blocks, COIN $\rightarrow$ LINK), most plausibly a liquidity/ session-timing effect rather than information flow. As with W , this is persistent lead-lag *association*; with a temporal ordering it is closer to Granger precedence than W , but still not structural causality.

BIC selects a negligible $\gamma=0.02$, which alone shrinks $\|\Delta_W\|_1$ by $\sim 63\%$). Where the synthetic ground truth had selection choosing $\gamma > 0$ in every seed, the real data has selection finding *no change graph worth encoding*, an orthogonal witness to the same null.

5 Discussion and limitations

The methodological contribution is validated on ground truth: a fused, Student- t DBN with an exact-prox ADMM solver recovers a sparse change graph substantially better than the DYNOTEARS-

equivalent, with the advantage associated primarily with the ADMM/exact-prox implementation, secondarily with the heavy-tailed loss and fusion, and robust to a distribution-free transform and to principled penalty selection. The solver ablation is a practical baseline comparison rather than a perfectly scale-matched prox-only isolation. The empirical study delivers a carefully-controlled negative result: scheduled macro events raise volatility (which the matched null absorbs) but do not measurably reorganize the contemporaneous conditional dependency graph at hourly RTH frequency over this window. Lagged edge-wise tests also find no stable changing edges; the lone global lagged-norm rejection is best read as an event-regime estimation artifact. Held-out selection independently declines to encode a change graph.

Several limitations bound these conclusions. (1) Frequency and timing: real cross-asset lead-lag plausibly lives at sub-hourly scales an hourly two-year panel cannot resolve, and 08:30-ET releases are observed only post-open, so a minute-scale, near-24h panel is the natural next test. (2) Event-regime size: with 103 event bars, fusion has little out-of-sample signal and exact support recovery is hard, which is exactly why ground-truth simulation, not held-out density, is the right yardstick. (3) Scope: the closed-form prox is $K=2$ only; $K > 2$, learned ν , and per-regime scales are deferred. (4) Identifiability: with free per-variable scales we make no equal-variance Gaussian identifiability claim [Peters and Bühlmann, 2014]; Student- t here is a robust M -estimator yielding more efficient recovery under heavy tails, as the synthetic results bear out.

6 Conclusion

We presented FR-tDBN, a fused-regime Student- t dynamic Bayesian network whose estimand is a sparse change graph, solved by consensus ADMM with a closed-form fused-lasso prox inside the NOTEARS augmented-Lagrangian loop. On a var-sortability-controlled benchmark it recovers the change graph far better than the DYNOTEARS-equivalent, with the gain associated mainly with the ADMM/exact-prox implementation. Applied to a complete cross-asset panel, it validates the contemporaneous heavy-tailed DAG but finds, under volatility-matched nulls and corroborating model selection, that scheduled macro events do not reorganize the contemporaneous dependency structure at hourly resolution; lagged edge-wise tests also find no stable event-specific edges. This is an honest, controlled null distinct from the validated method.

Reproducibility. All results come from one codebase (`frtddb/`); the synthetic grid/sweep, real-panel run, robustness, baselines, selection, and figures are produced by scripts in `scripts/`, with 100 unit tests passing. Data is hourly Yahoo Finance; the event calendar is verified against official BLS/Fed schedules. All code and data are available at <https://github.com/0xknxwledge/TTIC31180FinalProject>.

Acknowledgments. Drafting and figure code were assisted by Claude (Anthropic) and cross-checked by Codex; all results, analysis, and final content are the author’s.

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A Robustness of the recovery advantage to the choice of robust loss

Is FR-tDBN’s change-graph recovery advantage specific to the Student- t likelihood, or does any robust data term capture it? We add two alternative data terms and rerun the synthetic grid under the headline ADMM exact-prox solver with W -only fusion ($\lambda=0.03$, 24 fits/cell). The first is a convex *pseudo-Huber* loss $\ell_\delta(e) = \delta^2 (\sqrt{1 + (e/\delta\sigma)^2} - 1)$ ($\delta=1.345$ in robust-scale units; quadratic near zero, with linearly bounded influence in the tails), which is robust but not a normalized density and so is scored on recovery only. The second is a *Normal-Inverse-Gaussian* (NIG) likelihood: a Gaussian whose variance is drawn from an inverse-Gaussian (a normal variance–mean mixture; we fit the standardized symmetric case with tail-steepness $\alpha=1$). NIG is itself a proper heavy-tailed density, so unlike pseudo-Huber it could also join the held-out comparison; we report recovery for a like-for-like contrast.

data term	$\nu = 5$		$\nu = 3$	
	independent	fused	independent	fused
Gaussian	0.755 (.027)	0.794 (.027)	0.590 (.034)	0.625 (.029)
Student- t	0.869 (.023)	0.854 (.032)	0.684 (.032)	0.707 (.028)
pseudo-Huber	0.868 (.024)	0.846 (.030)	0.677 (.030)	0.683 (.026)
NIG	0.865 (.024)	0.871 (.029)	0.637 (.032)	0.691 (.027)

Table 5: Change- W AUROC under four data terms (ADMM solver, W -only fusion, mean (s.e.) over 24 fits/cell). Best per column in **bold**.

Three takeaways. First, all three robust losses beat Gaussian by a wide margin (+0.09 to +0.13 in the independent fits), so the recovery gain reflects heavy-tailed modeling broadly rather than any one parametric family. Second, at moderate tails ($\nu=5$) the three are statistically indistinguishable (0.85 to 0.87; the fused NIG is nominally best). Third, at the heaviest tails ($\nu=3$) the ranking tracks tail shape: the Student- t , whose polynomial tails match the data, leads, while NIG’s lighter (exponentially-tempered) tails at a fixed α trail in the independent fit (0.637) and largely recover under fusion (0.691), with pseudo-Huber in between. We therefore keep the Student- t as the headline: it is a proper likelihood (enabling the held-out comparisons of §4), its power-law tails match returns, and it is best or tied-best under the heaviest tails. The broader point is that the methodological advantage is a robustness effect shared across heavy-tailed losses, not an artifact of the Student- t form.